# MPLS operation experiences and challenges

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# What we achieved for

- Robust and Stable backbone
  - Some technique/gimmicks for stable routing
  - Traffic Engineering/Fast Reroute
- Business opportunity
  - Multi-service converged networking

# MPLS operation experiences

- Provisioning
  - Lots of vrf/interface
  - Too many te-lsp and bypass-lsp

# MPLS operation experiences

- Fault detection
  - Signaling protocol status
  - LSP ping
  - Routing convergence monitoring

# MPLS operation experiences

- Network Observation
  - MPLS related MIB
    - LSR-MIB
    - LDP-MIB
    - TE-MIB
    - FRR-MIB

# What MPLS is going to

- Full automation
  - Auto-mesh
  - Auto-optimization
- High coverage
  - Multi-area/domain
  - Multi-layer
- High density
  - All of multi-service customers on MPLS network

# What our challenges are

- Mapping network resources/event to services/customers
  - What does this LSP serve?
  - Which services/customers affected on this fail?
- Observing misbehavior
  - When an unexpected event occur, how do we detect and fix it?
- Management plane robustness
  - When we lost mgmt connectivity, network is in the dark